DataFlex: A Unified Framework for Data-Centric Dynamic Training of Large Language Models

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Research Questions

- How can data-centric training methods (data selection, mixture optimization, reweighting) be unified under a single framework to improve reproducibility and fair comparison?
- Can dynamic data optimization consistently outperform static training approaches across different model sizes and datasets?
- What system design enables scalable integration of diverse data-centric methods into existing LLM training pipelines?

Core Idea

Existing data-centric training methods (selecting which samples to train on, mixing data domains, reweighting samples) are scattered across isolated codebases with inconsistent interfaces, making comparison and deployment difficult. DataFlex solves this by creating a unified framework built on LLaMA-Factory that treats data as a first-class optimization variable. The key insight is providing three trainer abstractions (Select, Mix, Weight) with pluggable components that can dynamically control data usage during training while maintaining compatibility with existing infrastructure. This enables fair comparison of methods and practical deployment at scale.

Key Figure

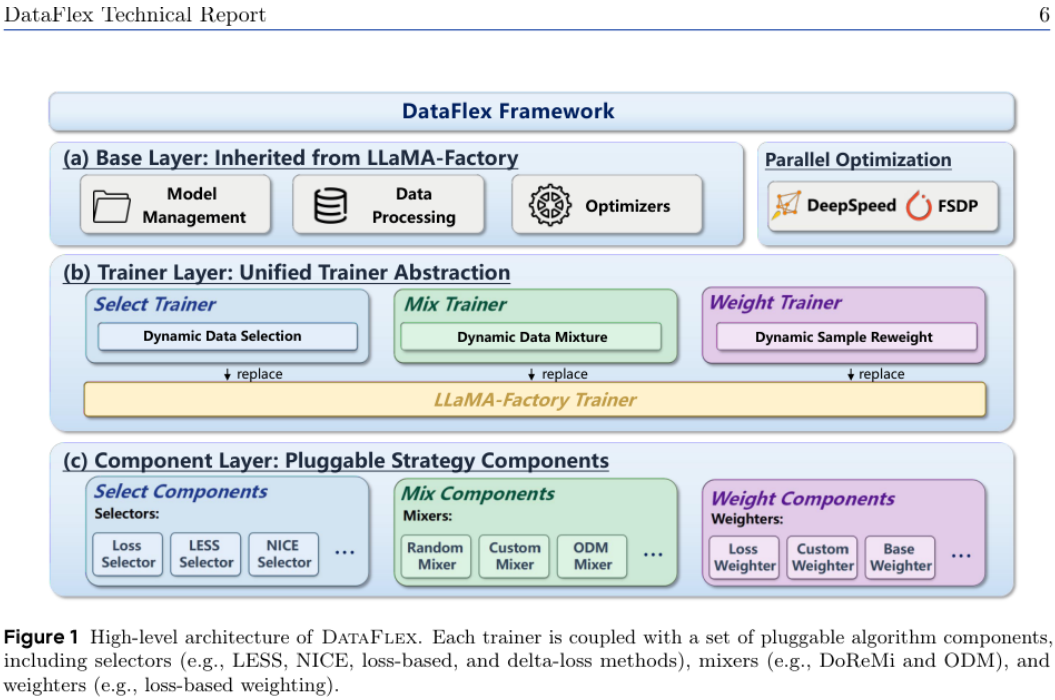

Figure 1 High-level architecture of DataFlex. Each trainer is coupled with a set of pluggable algorithm components, including selectors (e.g., LESS, NICE, loss-based, and delta-loss methods), mixers (e.g., DoReMi and ODM), and weighters (e.g., loss-based weighting).

Figure 1: Shows the three-layer architecture of DataFlex with base layer (LLaMA-Factory), trainer layer (Select/Mix/Weight trainers), and component layer (pluggable selectors/mixers/weighters), illustrating the modular design that enables unified data-centric training.

Limitations

- Framework is built specifically on LLaMA-Factory, potentially limiting compatibility with other training frameworks
- Gradient-based methods require full parameter gradients which may not be available under all model parallelism settings
- System overhead from dynamic data operations may impact training efficiency despite optimizations
- Limited to three paradigms (selection, mixture, reweighting) and may not cover all possible data-centric approaches